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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 06/01/2006 |
| **PI Name:** | Roux, Stanley Ph.D. | | |
| **Project Title:** | Cellular bases of light regulated gravity responses | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Austin | **State:** | TX |
| **Zip Code:** | 78712-1240 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Space Biology<br>01-OBPR-06 |
| **Start Date:** | 11/01/2002 | **End Date:** | 10/31/2007 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 9 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE through 10/31/2007, per A. Chu (9/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Clark, Gregory ( The University of Texas at Austin ) | | |
| **Grant/Contract No.:** | NAG2-1586 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The overall goal of this project is to clarify the cellular mechanisms that control light-regulated gravity responses. Results obtained this year advanced the overall objective of this project by providing new information on steps in the transduction chain for gravity-regulated responses in which Ca2+ is likely to exert a critical influence. Here we highlight two important advances that contributed toward achieving project objectives during the current funding year. These advances are: 1. Further characterization of expression of each of the 8 different genes in Arabidopsis that encode the calcium-binding annexin proteins. This advance allows us to use a rigorous genetic approach to evaluate the role of each of 8 different annexins in plant growth and development generally, and specifically in gravitropism. 2. The use of microarray technology to perform a global assay of changes in the abundance of mRNAs encoded by over 3200 different genes that occur in single germinating spores of Ceratopteris during the first 48 h of their gemination, including genes likely to regulate the calcium current that is directed by gravity in spores and appears to be necessary for subsequent gravity-directed developmental polarity. Novel data arising from these analyses identify genes likely to be critical for the germination and subsequent early development of diverse cells and tissues emerging from dormancy, including genes needed for gravity to direct the polarity of cell growth. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research does not directly seek to understand a disease or malady that affects humans on Earth and/or in space, nor does it seek to develop new therapeutics for alleviating symptoms of a malady on Earth. However, the findings will help to understand the growth of plants better, and since plants are a crucial source of food for humans, this research does seek to understand the malady of malnutrition. Also, this research does yield a new understanding of basic biological processes, specifically the processes of plant growth and the cellular mechanisms whereby gravity can affect the developmental polarity in cells. Further, this research points to a real role of gravity in regulating growth and development of plants on Earth and thus reveals potential problems in achieving normal growth and development of plants in space. Finally, the health of humans is inextricably linked to the ability to control and continuously improve the growth of plants. This, in turn, requires an improved understanding of the molecular mechanisms that control growth in plants. The accomplishments of this research contribute to that improved understanding, and thus indirectly benefit humans. |
| **Task Progress:** | During the past year there were two main areas of progress, one that employed the model flowering plant Arabidopsis, and one using the model single cell system for gravitational studies, Ceratopteris richardii.<br>In the Arabidopsis studies, our earlier work showed that annexins are highly expressed in secretory vesicles and, since gravitropism involves the asymmetric distribution of secretory vesicles to regions of cell expansion, annexins are preferentially distributed on the faster growing side of gravitropically curving tissues. Although this work implicates annexins in growth responses it does not specify which of the 8 annexin family members are most likely to be critical for growth.<br><br>To help answer this question we assessed the differential expression of annexins in different tissues or after different stimuli, using a sensitive method called quantitative real time reverse transcription PCR. We determined which annexins are expressed during germination and early seedling growth. Our results indicated that transcripts for all eight annexins are present in germinating seeds and that transcript levels for all the annexins increase by 7 days of normal growth. We assayed transcript levels in dark grown roots, cotyledons, and hypocotyls and found that the relative abundance of each annexin varied in these dark grown tissues, so different members of the annexin family are regulated differently in the same tissues. We also examined the effects of growth-altering irradiations of red and far red light on annexin expression in dark-grownseedlings, and found that growth-inhibiting levels of red light significantly altered transcript levels in hypocotyls and cotyledons for only two members of the gene family. Finally, we monitored annexin expression changes in response to a variety of abiotic stresses that typically suppress growth. We found that expression of most of the Arabidopsis annexin genes is differentially regulated by exposure to salt, drought, and high and low temperature conditions, indicating a likely role for members of this gene family in stress responses. These results allow us to evaluate better which annexins are most closely linked with growth control and lay the groundwork for gene knockout studies that will reveal which family members play critical roles in the control of gravitropic growth.<br><br>In the Ceratopteris studies, our earlier work had shown that the stimulus of gravity fixes the polarity of growth of single-celled spores of this fern within the first 16 hours after the spores are induced by light to germinate. However we had no information on what were the major gene expression changes that occurred during this critical period. To help answer this question we performed used microarray technology to perform a global assay of changes in the abundance of mRNAs encoded by over 3200 different genes that occur in single germinating spores of Ceratopteris during the first 48 h of their gemination.<br><br>We found that over 900 of the transcripts (29%) show a significant change between any of the five time points analyzed, and we annotated them based on their sequence similarity with the Arabidopsis proteome. Novel data arising from these analyses identify genes likely to be critical for the germination and subsequent early development of diverse cells and tissues emerging from dormancy, including genes needed for gravity to direct the polarity of cell growth. These findings will serve as a valuable base of information with which to compare the list of transcripts whose abundance is most dramatically altered by space flight, a list we are currently assembling using the same microarray technology. |
| **Bibliography Type:** | Description: (Last Updated: 10/16/2023) |
| **Abstracts for Journals and Proceedings** | Salmi M, Morris K, Porterfield DM, Roux SJ. "Microarray assessment of the effects of polarity-altering treatments on gene expression in developing Ceratopteris spores." Presented at annual Plant Biology meeting held in Seattle, WA, July 16-20, 2005.<br>Plant Biology 2005; Abstract No. 654, July 2005. , Jul-2005 |
| **Abstracts for Journals and Proceedings** | Clark GB, Cantero-Garcia A, Chou S, Roux SJ. "T-DNA mutational analysis and expression profiling of the Arabidopsis annexin gene family." Presented at the Annual meeting of Plant Biologists held in Seattle, WA, July 16-20, 2005.<br>Plant Biology 2005; Abstract No. 843, July 2005. , Jul-2005 |

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| **Abstracts for Journals and Proceedings** | Bushart TJ, Salmi ML, Roux SJ. "Studies of the gravity-directed calcium current in germinating spores of Ceratopteris richardii through CrACA1 and microarray analyses." Presented at the 21st Annual Meeting of the American Society for Gravitational and Space Biology, Reno, NV, November 1-4, 2005.<br>Gravitational and Space Biology. 2005 Nov;19(1):12, Abstract #29. , Nov-2005 |
| **Articles in Peer-reviewed Journals** | Salmi M, Bushart T, Stout S, Roux SJ. "Profile and analysis of gene expression changes during early development in germinating spores of Ceratopteris richardii." Plant Physiol. 2005 Jul;138(3):1734-45. PMID: 15965014 , Jul-2005 |
| **Articles in Peer-reviewed Journals** | Clark GB, Cantero-Garcia A, Butterfield T, Dauwalder M, Roux SJ. "Secretion as a key component of gravitropic growth: implications for annexin involvement in differential growth." Gravit Space Biol. 2005 Jun;18(2):113-4. PMID: 16044633 , Jun-2005 |
| **Articles in Peer-reviewed Journals** | Cantero A, Barthakur S, Bushart TJ, Chou S, Morgan RO, Fernandez MP, Clark GB, Roux SJ. "Expression profiling of the Arabidopsis annexin gene family during germination, de-etiolation and abiotic stress." Plant Physiol Biochem. 2006 Jan;44(1):13-24. PMID: 16531057 , Jan-2006 |
| **Articles in Peer-reviewed Journals** | Clark GB, Lee D, Dauwalder M, Roux SJ. "Immunolocalization and histochemical evidence for the association of two different Arabidopsis annexins with secretion during early seedling growth and development." Planta. 2005 Feb;220(4):621-31. PMID: 15368128 , Feb-2005 |